

PROJECT REPORT
StudySync - Secure Academic Resource and Communication Platform

Submitted By
Shwetanshu Bhatt
B.Tech Student
Graphic Era Hill University, Dehradun

Academic Year
2025-2026

Department
Computer Science and Engineering

1. Abstract

StudySync is a web-based academic resource and communication platform designed for students. The project is developed in multiple phases to ensure stability, scalability, and security.

2. Introduction

With the increasing use of digital technologies in education, students and teachers depend heavily on digital resources. Students often rely on social media groups and messaging applications, which are unstructured and lack security. StudySync is developed to solve these problems by providing a secure, organized, and role-based platform.

3. Problem Statement

The major problems faced in academic communication and resource management are:

- Lack of centralized note-sharing system
- Unorganized distribution of study materials
- Insecure messaging platforms
- No role-based access control
- Difficulty in monitoring academic activities
- Risk of data loss and misinformation

These issues reduce learning efficiency and compromise data privacy.

4. Objectives

The main objectives of the project are:

- To develop a centralized academic platform
- To organize notes by year, semester, and subject
- To provide secure real-time communication
- To implement role-based access control
- To ensure user data privacy
- To improve student-teacher collaboration
- To build a scalable and maintainable system

5. Scope of the Project

Phase 1 (Initial Phase - 20%)

- User registration and login
- Role management
- Notes upload and viewing
- Basic chat rooms
- Application deployment

Phase 2 (Advanced Phase)

- Personal direct messaging
- End-to-End Encryption
- Encrypted group chats
- Admin dashboard
- Backup system

Phase 3 (Future Phase)

- Mobile application
- AI-based summarization
- Analytics dashboard
- Offline access
- Video conferencing

6. Project Description

StudySync is a full-stack web application that enables educational institutions to manage a

Functional Modules

1. Authentication Module
2. User Management Module
3. Notes Management Module
4. Chat Module
5. Admin Control Module
6. Security Module

Working Procedure

1. Users register and log in.
2. System verifies user roles.
3. Teachers and admins upload notes.
4. Students browse and download notes.
5. Users communicate through chat rooms.
6. Messages are stored securely.
7. Encryption is added in later phases.

7. System Architecture

The system follows a client-server architecture.

Components

- Web Browser
- Frontend Application
- Backend Server
- Database Server
- Cloud Storage
- Socket Server

Architecture Flow

1. Client sends request to frontend.
2. Frontend communicates with backend APIs.
3. Backend processes requests.
4. Data is stored in MongoDB.
5. Files are stored in cloud storage.

6. Socket.io manages real-time chat.

Client-side encryption is implemented in Phase 2.

8. Technology Stack

Frontend

- React.js
- Tailwind CSS
- Axios
- Socket.io Client

Backend

- Node.js
- Express.js
- Socket.io

Database

- MongoDB (Mongoose)

Cloud Storage

- Cloudinary / Firebase Storage

Authentication

- JWT
- bcrypt

Security

- Web Crypto API
- AES and RSA Encryption
- IndexedDB

Deployment

- Vercel
- Render

9. Database Design

User Collection

userId name email password role year branch publicKey

Subject Collection

subjectId name year semester

Notes Collection

noteId title subjectId fileUrl uploadedBy createdAt

Chat Collection

messageId senderId roomId cipherText timestamp

10. Development Methodology

The Agile methodology is followed.

Agile Phases

1. Requirement Analysis
2. Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

The project is developed in small iterations with continuous testing.

11. Implementation Plan

Phase 1 (10 Days)

Day	Task
-----	-----
1	Setup
2	Authentication
3	Roles
4	Notes Backend
5	Notes Frontend
6	Chat Backend
7	Chat Frontend
8	Isolation
9	Testing

| 10 | Deployment |

12. Testing Strategy

Types of Testing

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

Tools

- Postman
- Jest
- Manual Testing

13. Possible Difficulties

1. Security vulnerabilities
2. Server overload
3. File upload failures
4. Database issues
5. Encryption complexity
6. Key management problems
7. Deployment errors

14. Solutions to Difficulties

- Password hashing
- Token expiration
- Rate limiting
- Database indexing
- Cloud storage
- Secure key storage
- Proper configuration

15. Advantages (Pros)

- Centralized platform
- Secure communication
- Organized resources
- Role-based access
- Scalable system
- Industry-relevant project
- Improves productivity

16. Limitations (Cons)

- Requires internet
- High complexity
- No offline mode
- Encryption overhead
- Cloud dependency
- No mobile app initially

17. Applications

- Colleges
- Universities
- Coaching centers
- Training institutes
- Online learning platforms

18. Security Considerations

- Hashed passwords
- Token authentication
- Role validation
- HTTPS encryption
- Client-side encryption
- Secure storage

19. Ethical and Legal Considerations

- User privacy protection

- Data security compliance
- No unauthorized monitoring
- Responsible data usage
- Transparent policies

20. Future Enhancements

- End-to-End Encryption
- AI summaries
- Mobile application
- Offline access
- Analytics
- Video conferencing
- Auto backups

21. Project Outcome

The project delivers a secure academic platform that organizes resources and communication e

22. Summary

StudySync is a scalable and secure academic platform that improves learning collaboration. 1

23. Conclusion

The system successfully addresses problems related to unorganized academic resources and ins

24. References

1. React.js Documentation
2. Node.js Documentation
3. MongoDB Documentation
4. Socket.io Documentation
5. Web Crypto API Documentation
6. JWT Authentication Guide
